GARDA: Granular Association-Rule-Based Data Imputation Approach for IoT Sensor Networks

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***Abstract*—Data imputation is the process of replacing missing data with substituted values to maintain the integrity and usability of a dataset. It is critically important for improving the analysis of data collected from Internet of Things (IoT) sensors and devices, which are often affected by failures or communication issues that can result in data loss. The absence of missing data in IoT applications is of paramount relevance to avoid errors, accidents, and wrong decisions. This work introduces—granular association-rule-based data imputation approach (GARDA)—a data imputation method based on association rules and granular computing for imputing missing data coming from IoT sensor networks characterized by sensor data with high correlation. GARDA performs the granulation of continuous data with interval-valued sets from which granular association rules are mined. These association rules are used to impute the missing intervals. Finally, the intervals are transformed back to continuous data to impute missing values. The approach has been evaluated on the Intel Lab and Beijing Air datasets and compared with state-of-the-art techniques, demonstrating good performance.**

***Index Terms*—Association rule, data imputation, granular computing (GrC), missing value.**

I. Introduction

IN THE context of Internet of Things (IoT), the problem of data imputation is critical [1], [2]. Data imputation can be a useful and fundamental step in all those IoT applications and systems that use data collected by a sensor network [3], [4], as it is quite frequent to have to deal with missing data, contradictory data, or low-quality data [5], [6]. In particular, IoT sensors and their communication networks are inherently characterized by low processing power, low storage capacity, limited battery power, and limited communication bandwidth [7], [8]. These constraints may degrade the quality of data collected during the perception phase, with a negative impact on the application that is connected to these sensor networks [9], [10].

Received 17 July 2025; revised 12 December 2025 and 26 March 2026; accepted 11 June 2026. Date of publication 17 June 2026; date of current version 26 August 2026. *(Corresponding author: Giuseppe D'Aniello.)*

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Digital Object Identifier 10.1109/JIOT.2026.3704862

Data imputation also plays a fundamental role in the context of machine learning systems. When we have to use a machine learning model, we have to train it with huge and high-quality datasets. These datasets should contain few missing values; otherwise, the training is counterproductive, leading to the so-called garbage-in–garbage-out phenomenon [11]. In practical, real-world contexts, it is difficult to find datasets that are not affected by error and missingness [12]. One of the possibilities to handle this problem is to delete the records of the dataset containing missing data. However, more sophisticated data imputation techniques have been proved to improve the performance of machine learning tasks. Traditional imputation algorithms, such as mean/median/mode imputation, linear regression, and k-means imputation, can be used to recover data in the aforementioned situations [13]. However, in some cases, especially when the missing rate is high, these techniques may have difficulties in achieving satisfactory performances. Although deep-learning-based approaches often provide better performances than traditional methods, in edge computing scenarios, increasingly applied in IoT, devices frequently lack the storage capacity and computational power to run such models [14]. In these contexts, it is necessary to adopt data imputation approaches that can compensate for missing data in small networks or subnetworks, enabling data cleaning even under limited computational resources [15]. Furthermore, the use of traditional nondeep methods is now encouraged due to the enormous environmental damage that deep learning is causing [16]. Literature is, in fact, focusing on sustainable AI techniques that reduce the environmental impact [17].

To address these challenges, this work proposes—the granular association-rule-based data imputation approach (GARDA)—a novel, lightweight data imputation approach based on association rules and on the principles of granular computing (GrC) designed for IoT devices with limited power and battery, and for small, highly correlated sensor networks. GrC is a computing paradigm useful to handle uncertain data [18], [19], [20]. A key novelty of the approach lies in its ability to exploit association rules even in the case of continuous data through the introduction of granular association rules. Specifically, the data are first transformed into granules; subsequently, association rules are mined on this granulated representation. Finally, the imputed granules are reconstructed to obtain continuous values. The approach is enhanced by employing a hybrid algorithm, in which values that would be

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imputed by weak association rules are instead imputed using a traditional statistical method, such as linear regression (LR). This results in improved performance compared to using either granular association rules or the traditional method alone, while being considerably less resource-intensive than state-of-the-art deep learning models.

The article is organized in the following way. Section II reviews the state-of-the-art data imputation techniques. Section III explains the hybrid granular association-rule data imputation technique we designed. Section IV describes the dataset, the evaluation methodology, and the results obtained. Section V concludes the work.

II. Related Works

Data imputation is the process of reconstructing missing data in a dataset [21], [22]. The missingness (i.e., the probability that a datum is missing) is generally classified into: 1) missing completely at random (MCAR), when the missingness is unrelated to its or any other missing or observed values [23]; 2) missing at random (MAR), when the missingness is related to the observed data but not to the missing ones [24]; and 3) missing not at random (MNAR), when the missingness is related to the missing data themselves [25].

In what follows, we briefly review some traditional association-rule and neural network (NN)-based imputation techniques, together with state-of-the-art granular association-rule imputation techniques.

A. Traditional Imputation Techniques

Traditional imputation methods rely on statistical measures, including the mean, mode, and median of observed values. However, these approaches often prove inadequate for MCAR data, especially when the missingness rate exceeds 10% or when internal data correlation is high [26], [27]. Other traditional imputation methods include regression-based techniques, such as LR [28], polynomial regression [29], singular value decomposition (SVD) [30], as well as clustering-based approaches like K-neighrest neighbor (KNN) [31], K-means [32] or fuzzy C-means (FCM) [33].

B. Association-Rule Imputation

Association-rule mining is a family of rule mining techniques concerning the identification of recurrent patterns among various items in a dataset. Algorithms at the state of the art in association-rule mining include a priori [34], a priori TID [35], a priori hybrid [36], and FP-Growth [37].

Some data-imputation techniques use association-rule mining to reconstruct missing values by exploiting frequent patterns. Wu et al. [38] proposed a categorical data imputation method by ranking association rules. Wu et al. [39] presented a method that uses association rules to impute missing data in a given database based on a weighted voting strategy. Jiang [40] developed an approach based on mining closed frequent itemsets; these provide a condensed, yet complete representation of data patterns by eliminating redundancy. Chaurasia [41] proposed a data imputation technique based on multiple imputation combined with rules based on univariate F- and Beta-statistics from multiple imputed data. The advantage of this approach is that univariate F- and Beta-statistics are efficient from a computational point of view while providing a good inferential capability. Hong and Wu [42] introduced an iterative missing-value completion method based on the support values of robust association rules (RARs) to extract useful association rules for inferring missing data. The previous techniques are designed and applied only on categorical data. D'Aniello et al. [9] extended this approach to accommodate IoT datasets containing continuous values. Their work proposes a quality-aware sensor data management framework, based on virtual sensors, in which the quality requirements are chosen by system users. Our article builds on the algorithm proposed by [42], which was extended to continuous values in [9], and integrates these approaches with the GrC paradigm. Most importantly, the proposed integration avoids the manual discretization of continuous values by representing them with interval sets.

C. NN-Based Imputation

NNs can be used to learn hidden patterns and relationships within the data to predict and impute missing values. Among these methods, many are focused on multivariate time-series data, typical of IoT applications. These can be divided into two main categories: recurrent-NN-based (RNN) and self-attention-based. M-RNN [43] belongs to the first category and uses a multidirectional recurrent deep NN composed of an interpolation block that exploits the correlation within each data stream, and an imputation block that exploits the correlation across data streams. SAITS [44] is self-attention-based and uses a weighted combination of two diagonally masked self-attention (DMSA) blocks to learn the temporal and feature correlations between the measurements.

D. Granular Imputation

Data imputation is characterized by uncertainty because when a piece of information is lost, it is impossible to recover it with certainty. The intrinsic capability of dealing with uncertain and noisy data of GrC makes it ideal for working on data imputation.

GRC has emerged as a conceptual and algorithmic platform aimed at the representation and processing of information granules [45], [46]. Granules are the fundamental building blocks of GrC, where a granule is any subset, class, object, cluster, or element of a universe that can be seen as a more specific and restricted part of it [47]. Granulation aims at the composition and decomposition of granules [48]: they can be fuzzy sets, shadow sets, rough sets, and interval sets. In this article, we will use the interval set [49], in which an element is part of or not of a granule if its value is contained within the boundaries of the interval representing the granule. Along with the set theory comes a well-developed discipline of interval analysis [50]. According to [46], an interval set A is described in the following way:

$$A(x) = \begin{cases} 1, \ if \ x \in A \\ 0, \ if \ x \notin A \end{cases} \quad (1)$$

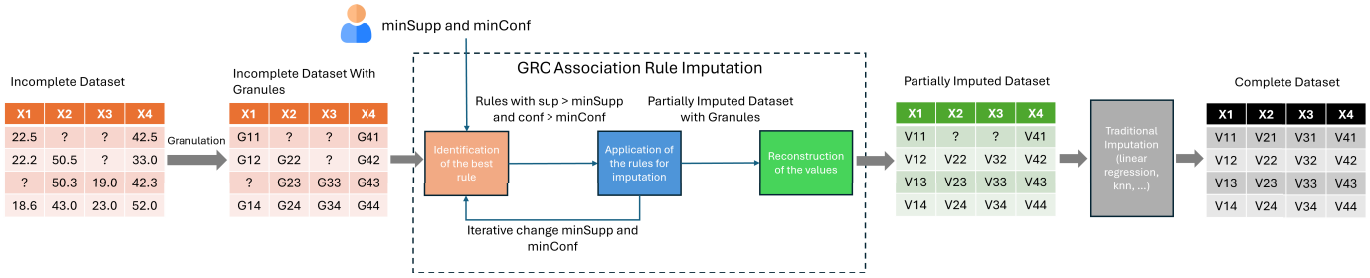


Fig. 1. GARDA architecture.

where $A(x)$ stands for a value of the characteristic function of set A at point x.

In particular, Zhong et al. [51] established a two-stage imputation mechanism, in which they started with methods of numeric imputation and then formed a granular representation of missing values. They proposed two imputation strategies. In the first, imputation is realized for individual variables of datasets and subsequently enhanced by the buildup of information granules. The second approach adopted fuzzy clustering, specifically FCM, to establish a structure in the data and then leverage this information in the imputation process.

Hu et al. [20] proposed a novel information-granule-based classification method for incomplete data. Their method first identifies the structural backbone of labeled subspaces via fuzzy clustering; subsequently, it develops specialized information granules to impute missing values according to refined prototypes and classification results.

Hu et al. [19] introduced a novel and straightforward approach that creates information granules to represent missing data and build granular fuzzy models directly from resulting hybrid granular and numeric data. In the evaluation, they rely on the principle of justifiable granularity.

III. Mining Granular Association Rules for Missing Data Imputation

The proposed GARDA is illustrated in Fig. 1. GARDA is composed of three phases: 1) a granulation approach for the discretization of continuous values inside the dataset with the missing data to impute; 2) granular association rules to partially impute the missing values by identifying the intervals in which these values may exist; and 3) a traditional data imputation technique (e.g., LR) to fill in the remaining missing values. Each of these three phases is described in the pseudocode of Algorithm 1. In particular, Algorithm 1 can be interpreted in three blocks. Lines 1–6 perform the granulation of continuous attributes, Lines 7–84 describe the granular association-rule-based imputation procedure, and Lines 85–95 reconstruct continuous values from the imputed granules and apply a traditional imputation method only to the values that remain missing.

The rationale for this hybrid approach is that the granular-rule-based approach—based on the discrete imputation approach proposed by Hong and Wu [42]—is highly effective when leveraging high-confidence rules. Conversely, the imputation performance degrades with low-confidence

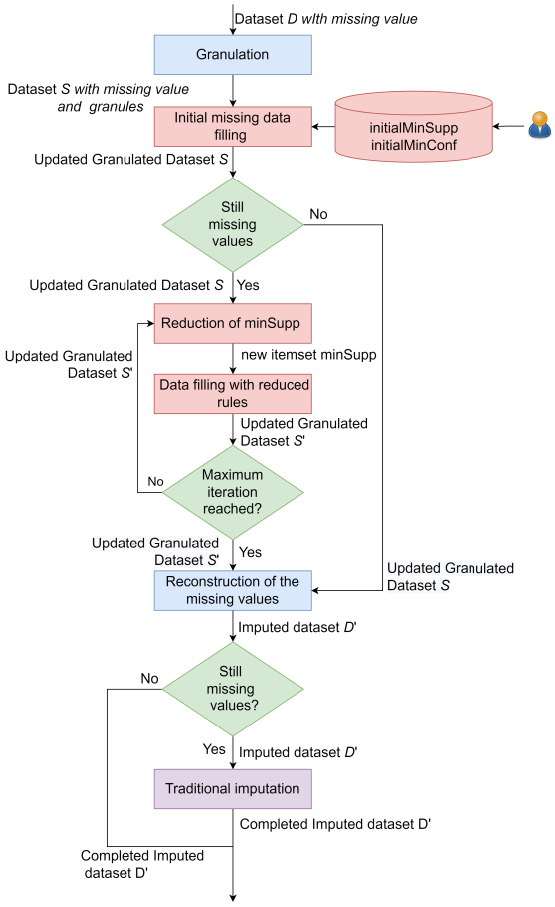


Fig. 2. GARDA flow diagram.

rules, yielding worse results than traditional data imputation techniques. However, the latter are outperformed by the rule-based method on cells in which it imputes with high confidence. Therefore, we decided to propose a hybrid approach that can capture the best of both techniques. In addition, after part of the missing cells have been effectively imputed by the rule-based approach, the imputation performed by the traditional algorithm is enhanced. These considerations are supported by experimental evidence, which will be shown in Section IV and by the results in Tables I and II.

The GARDA approach, illustrated in Fig. 2, begins with a dataset, represented as a matrix, in which some cells are empty. The missing values in these empty cells are reconstructed

Algorithm 1 GARDA

Input IncompleteDataset, initSup, initConf, numMaxIter, rSup, rConf
Output ImputedDataset

▷**Granulation**
if *IncompleteDataset* have continuous values **then**
  IncompleteDiscreteDataset ← *Granulation(IncompleteDataset, Discretization Parameters, MinMaxPerAttributes)*
else
  IncompleteDiscreteDataset ← *IncompleteDataset*
end if
▷**GrC Association-rule-based Imputation**
rows ← *getRows(IncompleteDiscreteDataset)*
Step 1:
for all *row* in *rows* **do**
  for all *item* in *row* **do**
    if *support(item)* ≥ *initSup* **then**
      L1.add(item)
    end if
  end for
end for
Step 2:
L2 ← *get2LargeItemSets(L1)*
Lk.addAll(L2)
do
  k++
  LkList.addAll(Lk)
  createKplus1Itemset(Lk, CkPlus1, LkPlus1, k)
  Lk ← *LkPlus1*
while *size(LkPlus1)* > 0
Step 3:
for all *associationRule* from itemsets in *LkList* **do**
  if *Conf(associationRule)* > *initConf* **then**
    add *associationRule* to *associationRules*
  end if
end for
Step 4:
for all *row* in *rows* **do**
  for all *item* in *row* **do**
    if only an association rule is found **then**
      use such rule
    end if
    if more association rules are found **then**
      Use the rule with maximum support or confidence value
    end if
    if more association rules are found with the same support and confidence **then**
      leave the value missing
    end if
  end for
end for
Step 5 and 6:
$y \leftarrow 1$
while *missingValuesInDataSet* and $y <$ *numMaxIter* and $minSup > 0$ **do**
  Step 7:
  $minSup \leftarrow \frac{initSup}{rSup \cdot y}$
  $minConf \leftarrow \frac{initConf}{rConf}$
  Step 8:
  for all *row* in *rows* **do**
    for all *missingValue* in *missingValues* **do**
      for all *possibleValue* in *possibleValues* **do**
        add itemset(missingItem, possibleValue) to C2b
      end for
    end for
  end for
  Step 9:
  for all *itemset* in *C2b* **do**
    if $itemsetSupport \geq minSup$ **then**
      add itemset to L2b
    end if
    add itemset(missingItem, possibleValue) to C2b
  end for
  Step 10:
  $reducedAssociationRules \leftarrow addRulesFromItemSets(L2b, C2b, minConf)$
  Step 11 and 12:
  for all *row* in *rows* **do**
    for all *item* in *row* **do**
      if only an association rule is found **then**
        use such rule
      end if
      if more association rules are found **then**
        Use the rule with maximum support or confidence value
      end if
      if more association rules are found with the same support and confidence **then**
        leave the value missing
      end if
    end for
  end for
  y++
end while
▷**Reconstruction and Traditional Imputation**
if *IncompleteDataset* had continuous values **then**
  PartiallyFilledDataset ← *ValuesReconstruction(IncompletedDiscreteDataset)*
else
  PartiallyFilledDataset ← *IncompletedDiscreteDataset*
end if
if *PartiallyFilledDataset* still have missing data **then**
  CompleteDataset ← *TraditionalImputation(PartiallyFilledDataset)*
else
  CompleteDataset ← *PartiallyFilledDataset*
end if

using the recurrent patterns among the dataset attributes, leveraging the correlation among the values.

The dataset is processed as shown in Fig. 1 by the following three phases.

1) *Granulation:* The process of discretizing continuous values. The values are grouped into granules whose characteristics depend on the granulation approach used.
2) *GrC Association-Rule-Based Imputation:* It takes in input an incomplete granular dataset, where each value is a granule and is characterized by missing data. The approach uses only the more consolidated and confident rules extracted from the patterns. The rules are applied to fill a subset of the originally missing values in the dataset. Some of the missing values in the dataset are filled with granules according to the exploited rules. The imputed granules will be used to recover the numeric values.
3) *Traditional Imputation:* The dataset obtained from the previous phase is given to a traditional imputation algorithm (e.g., LR) to fill the eventually remaining missing data from the previous imputation phase.

A. Formal Definition of the Approach

Let $\chi = \{x_1, x_2, x_3, \ldots, x_n\}$ be a set of variables with $x_i \in \mathbb{R}$ and $1 \leq i \leq n$, and $\mathcal{D}$ be a dataset which is a multiset of a subset of χ. We define a family of granulation functions $\Gamma_i(x_i, k_i)$ in which the specific function applied depends on x_i and k_i is the maximum number of granules relative to x_i. In particular, the function $\Gamma_i(x_i, k_i) : x_i \in \chi \rightarrow \{g_{ij}\}$, with $1 \leq j \leq k_i$, will output k_i granules for each x_i. In this way, each value of x_i is associated with granule g_{ij}.

Let $\mathcal{I} = \{g_{11}, g_{12}, \ldots, g_{21}, g_{22}, \ldots, g_{ij}\}$ be the set of granules generated, called items. A p-itemset X_p is a subset of the set of granules I containing p items, that is, $X_p \subseteq \mathcal{I}$ with $p = |X_p|$. Considering S a multiset of subsets (a dataset of already formed granules) of $\mathcal{I}$, we define $\sigma(X_p)$ as the transaction set for an itemset X_p as follows:

$$\sigma\left(X_p\right) = \left\{T \,\middle|\, X_p \subseteq T, T \in S\right\} \tag{2}$$

where T represents a single transaction in the dataset S.

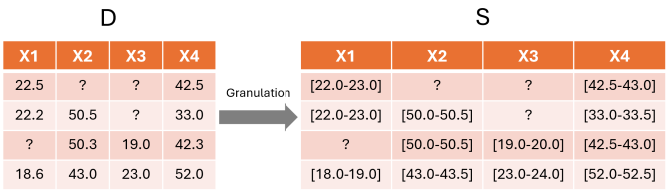
Considering two generic itemsets X_p and $Y_{p'}$ with p and p' the cardinalities of the itemsets (where typically $p' = 1$), an association rule can be defined as an expression $X_p \rightarrow Y_{p'}$ with $X_p \cap Y_{p'} = \varnothing$.

If the set S contains missing values, it is convenient to exploit the concept of RAR proposed by Ragel and Crémilleux [52], [53]. The RAR approach is designed to mine association rules in an incomplete dataset. Let $Dis(X_p)$ be the set of the disabled (missing) data of the itemset X_p

$$Dis(X_p) = \left\{T \,\middle|\, \exists A \in X_p\ , A = ?, X_p \subseteq T, T \in S\right\} \tag{3}$$

where ? is the missing value and A is the missing attribute belonging to X_p. The RAR support for an itemset X_p can be defined as

$$Sup(X_p) = \frac{\left|\sigma(X_p)\right|}{|S| - |Dis(X_p)|}. \tag{4}$$



X1	X2	X3	X4
22.5	?	?	42.5
22.2	50.5	?	33.0
?	50.3	19.0	42.3
18.6	43.0	23.0	52.0

X1	X2	X3	X4
[22.0-23.0]	?	?	[42.5-43.0]
[22.0-23.0]	[50.0-50.5]	?	[33.0-33.5]
?	[50.0-50.5]	[19.0-20.0]	[42.5-43.0]
[18.0-19.0]	[43.0-43.5]	[23.0-24.0]	[52.0-52.5]

Fig. 3. Application of the granulation approach to an incomplete dataset.

The confidence for an association rule $X_p \rightarrow Y_{p'}$ based on the RAR approach is defined as follows:

$$Conf(X_p \rightarrow Y_{p'}) = \frac{\left|\sigma\left(X_p \cup Y_{p'}\right)\right|}{|\sigma(X_p)| - \left|Dis(Y_{p'}) \cap \sigma(X_p)\right|}. \tag{5}$$

Based on such definitions, we design GARDA, with its flow diagram depicted in Fig. 2. Each phase of the approach is described in Sections III-B–III-D.

B. Granulation Approach

GARDA starts from an incomplete dataset to fill all the missing values, which can be discrete or continuous. An automatic granulation approach is used to create the information granules, thus avoiding the need for a manual discretization approach as instead proposed, for instance, in [9]. Obviously, if values are already discrete, there is no need for the granulation process.

In this section, we define three types of unsupervised granulation approaches that have been tested in this work and implemented in GARDA: equal-width [54], a static method; K-means [55], a clustering-based approach; and kernel density estimation (KDE) [56], a probabilistic method. Fig. 3 illustrates an example of a granulation approach based on equal-width intervals [54]. These approaches define the granulation functions to be applied for each type of attribute. We must define a granulation function for each variable that needs to be discretized.

1) Equal-Width: The granulation approach considers the minimal value a_{i0} and the maximal value a_{ik} of the variable x_i in D and divides the range into k_i equal-width discrete intervals. The approach calculates the interval width λ_i

$$\lambda_i = \frac{(a_{ik} - a_{i0})}{k_i} \tag{6}$$

and then granulates the numerical data into k_i discrete intervals bounded: $\{[a_{i0}, a_{i1}), \ldots, [a_{i(j-1)}, a_{ij}), \ldots [a_{i(k-1)}, a_{ik}]\}$

$$a_{ij} = a_{i0} + j\lambda_i \tag{7}$$

where $j = 1, 2, \ldots, k$.

In practice, the granules are created using the minimum and maximum values that each variable in the dataset may assume, along with a specified granule size. With this information, it is then possible to divide the range of possible values into equal-width intervals. If outliers are present, they are included in the granules at the edges of the range.

2) K-Means: Another granulation approach implemented in GARDA is K-means clustering [55]. In this approach, discretization emerges from grouping similar numerical values rather than uniformly partitioning the domain. The method applies the K-means algorithm to the attribute values of x_i, specifying the desired number of clusters k_i. Once the centroids $c_1, \ldots, c_{k_i}$ have been computed, each value is assigned to the nearest cluster.

For each resulting cluster C_j, the corresponding discretized interval is defined as

$$I_j = \left[\min\left(x_i \in C_j\right),\ \max\left(x_i \in C_j\right)\right].$$

This procedure captures the structure of the data: intervals are narrower in dense regions and wider in sparsely populated ones. Consequently, K-means-based granulation is useful when attribute distributions are nonuniform, multimodal, or naturally clustered, allowing the discretization to adapt to the patterns rather than relying only on domain-based spacing.

3) KDE: KDE is another granulation approach introduced in [56]. The method is based on adapting interval boundaries using the estimated probability density of the data. Unlike equal-width or equal-frequency discretization, KDE-based granulation identifies regions where the density changes significantly and places cuts accordingly.

For each candidate cut-point T, the method compares: 1) the density induced by the split using simple binning and 2) the density estimated through a kernel density estimator.

A score is computed for each candidate cut-point as

$$\text{Score}(T) = \sum_i \left(p\left(x_i\right) - f\left(x_i\right)\right)$$

where $f(x_i)$ is the binned density and $p(x_i)$ is the kernel density estimated at x_i. Cut-points with the largest score highlight intervals that poorly fit the true density and should therefore be split.

The splitting process proceeds iteratively, always refining the intervals showing the worst correspondence between induced and estimated density. The optimal number of intervals is determined through a stopping criterion based on cross-validated log-likelihood. As a result, this granulation method does not require any user-defined hyperparameters.

Fig. 3 provides an example of the granulation mechanism, in this case using equal-width. In the starting D dataset, x_1 and x_3 are elements of the same type of attribute (e.g., temperature), while x_2 and x_4 are elements of another type of attribute (e.g., humidity). In this example, for the attributes x_1 and x_3, an interval width $\lambda_{x_1} = \lambda_{x_3} = 1$, while for x_2 and x_4, the interval width is $\lambda_{x_2} = \lambda_{x_4} = 0.5$. The granulation process produces the granular dataset S.

Once the granulation approach has replaced the nonmissing values with their corresponding granules, missing data are imputed using granular association rules. Missing values for which no rules with sufficient confidence are found are left as missing. At this point, the algorithm returns a dataset that is still incomplete but with values grouped in granules, and for this reason discretized.

C. GrC Association-Rule-Based Imputation

The imputation of the missing granular intervals is performed using the granular association rules, following the procedure reported in Algorithm 1, which also describes the pseudocode of all the steps of the GrC association-rule-based imputation approach presented in this section.

Initially, the granular association rules, mined from the original incomplete dataset, are used to roughly complete missing values in Steps 1–4. During this first pass (lines 9–45 of the Algorithm 1), GARDA extracts frequent itemsets from the incomplete granular dataset using the RAR-support definition and then derives the candidate granular association rules whose confidence is above the initial threshold. The input of the algorithm is the dataset S, where each element is an interval-valued granule I_i. The mining of the rules is performed considering the initial minimum support ($initSup$) and initial minimum confidence ($initConf$) chosen by the user, according to the following steps.

1) Find the RAR-support values of all the 1-itemsets. If the support of a 1-itemset X_1 is not lower than the threshold, put it in the set of frequent 1-itemsets, L_1.
2) Iteratively, find the other frequent itemsets with more than one item in an a priori-like way using the RAR-support evaluation. The set of frequent k-itemsets is called L_k.
3) Find the confidence value of each possible candidate granular association rule generated from the frequent itemsets. If the confidence of such a rule is not smaller than the threshold $minConf$, it is inserted in the set of the granular association rules, GAR.
4) Use the set GAR to infer the missing values of the incomplete dataset with the following substeps.
    a) If there is only one granular association rule that can be used to derive the missing value of an attribute in a tuple, then use the rule.
    b) If there is more than one granular association rule which can be used (i.e., multiple matches with the antecedents of such rules), then use the one with the maximum RAR confidence value; if more than one rule has the same maximum RAR confidence values, then use the one with the maximum RAR-support value; otherwise, keep the value as missing. The updated dataset is indicated with S'.

If the dataset is still incomplete at this stage, the algorithm enters an iterative phase where the minimum support is progressively decreased, and more association rules are found, as long as they satisfy the minimum confidence threshold, to impute as many values as possible. The iterations continue until the dataset is complete, the minimum support and confidence become too low, or a maximum number of iterations is reached.

Therefore, this second stage (lines 46–84) acts as an iterative recovery phase: after the initial high-confidence rule-mining pass, the algorithm progressively relaxes the thresholds to recover additional missing values that could not be imputed reliably in the first pass.

5) Check the dataset; if there are still missing values in the dataset, continue with Step 6; otherwise, simply execute the reconstruction of the values, which is visible in Fig. 2.
6) Let $y = 1$, where y is used to control the reduced minimum support threshold.
7) Set the reduced minimum support threshold $minSup$, and the reduced minimum confidence threshold $minConf$, where $rSup$ and $rConf$ are the reduction coefficients for support and confidence. The parameter y controls the progressive relaxation across iterations, so that additional rules are considered only after the initial, more selective pass. By doing so, weaker but still potentially useful dependencies are introduced gradually rather than from the start.
8) For each tuple with missing values, find the set of originally nonmissing attribute-value pairs in the tuple from the granular dataset and create the candidate 2-itemsets, with one item from the set of nonmissing attribute-value pairs and the other from the possible values in a missing value.
9) Find the RAR-support values of all the candidate 2-itemsets from the transformed granulated dataset S; if the RAR-support of a candidate 2-itemset X_2 is not smaller than the reduced support threshold $minSup$, put it in the set of reduced frequent 2-itemsets, L2'.
10) Find the confidence value of each possible candidate association rule generated from the frequent 2-itemsets, L_2', generated in Step 9. If the confidence of a candidate granular association rule is not less than the reduced confidence threshold, put it in the set of reduced granular association rules, $GAR.'$
11) Use the set of granular association rules GAR' to infer the missing values of the updated dataset S' with the following substeps.
    a) If there is only one granular association rule that can be used to derive the missing value of an attribute in a tuple, then use the rule.
    b) If there is more than one granular association rule that can be used to derive the missing value of an attribute in a tuple, then use the one with the maximum RAR confidence value. If more than one rule has the same maximum RAR-confidence value, then use the one with the maximum RAR-support value. If the maximum RAR-support values are still the same and the rules infer different values, then keep the missing value still unknown.
12) $y = y+1$ and repeat Steps 7–11 until there are no missing values in the updated dataset or y reaches a predefined $numMaxIter$ value.

D. Value Reconstruction and Traditional Imputation

At this point, after using the association rules to identify granules instead of missing values in the initial dataset D, the algorithm replaces the granule with a continuous value. In the case of K-means, the continuous value is given by the corresponding cluster centroid, whereas for equal-width and KDE, the reconstruction function used is the mean value between the two boundaries of the interval in the following way:

$$rec_i(g_{ij}) = \frac{endInterval_{ij} + startInterval_{ij}}{2}. \tag{8}$$

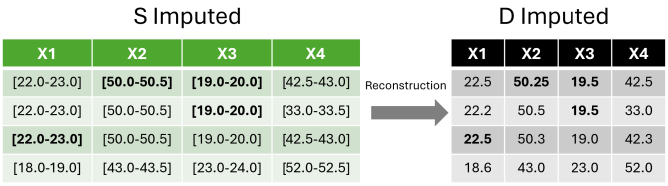


Fig. 4. Dataset reconstructed with the mean of the intervals. The initial missing values reconstructed by the algorithm are in bold.

An example of the reconstruction process is synthesized in Fig. 4.

Naturally, the values originally present in the dataset that were not missing are reconstructed with their original value.

If at the end of this phase, the dataset still has some missing values, a traditional imputation approach is used to complete the dataset. Only the entries that remain unresolved after the granular rule-based phase and the reconstruction step are passed to the traditional imputation module. This approach will be able to complete all missing values for which the confidence and quality of imputation of the rules would have been too inaccurate, including cases in which the row within the original dataset was completely empty, a case that the rule-based approach cannot handle, having no rules to apply. In contrast, traditional approaches can solve this problem, completing the imputation, and also gaining a performance advantage considering the reduced amount of missing values.

E. Computational Complexity

The computational complexity of the proposed imputation method is mainly determined by frequent itemset mining and association rule extraction, both based on an a priori-like strategy. Let k denote the number of items and n the number of rows. In the worst case, frequent itemset generation and antecedent–consequent enumeration both require exponential time, that is, $O(2^k)$, although in practice this cost is reduced by a priori pruning based on the minimum support threshold. Rule application further requires checking admissible imputations for missing entries, yielding an overall complexity of $O(n \cdot k \cdot 2^k)$ in the main phase. The iterative expansion phase relies on evaluating candidate 2-itemsets under progressively relaxed support and confidence thresholds, with complexity $O(n \cdot k^2)$. To the best of our knowledge, association-rule-based imputation methods for incomplete data are necessarily a priori-like, since more efficient alternatives, such as FP-growth [37], require complete transactions to ensure reliable frequency counting and valid structural compression. Therefore, no lower-complexity data-mining method currently appears to be directly applicable to missing data imputation in this setting [57].

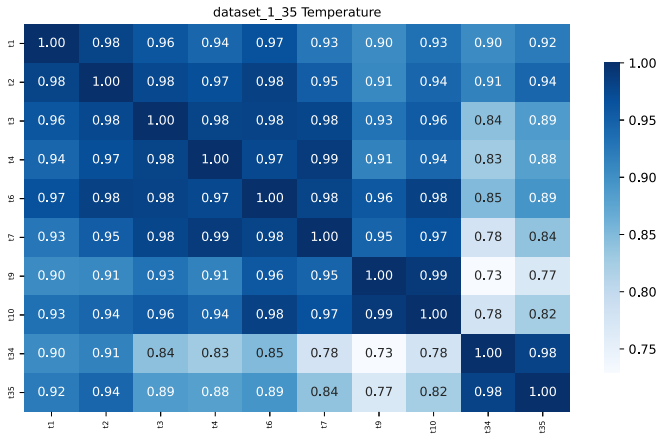


Fig. 5. Correlation matrix of the ten sensors considered in the Intel Lab.

IV. Evaluation

In this section, we describe the evaluation of the GARDA approach. Specifically, GARDA is compared with traditional data imputation techniques, including mean, median, mode, LR, KNN, K-means, and SVD-based, using several metrics like RMSE, MAE, NRMSE, and MAPE. In Addition, it is also compared with two state-of-the-art NN-based approaches: SAITS [44] and M-RNN [43], both introduced in Section II.

A. Dataset

The experiments for the data imputation techniques were conducted on the Intel Lab Dataset, which contains information about data collected from 54 sensors deployed in the Intel Berkeley Research Lab (https://db.csail.mit.edu/labdata/labdata.html) between February 28th and April 5th, 2004. Each entry was from a Mica2Dot sensor, which measures temperature, humidity, light, and battery voltage. A measurement provided by a sensor has the following format: date, time, epoch, mote id, temperature, humidity, light, and voltage. The sensors took a measurement approximately once every 31 s. The dataset includes a log of about 2.3 million readings from the 54 sensors installed in the lab. Epoch is a monotonically increasing sequence number from each measurement. Two readings from the same epoch number were produced from different motes at the same time, and there are also many missing epochs. Considering that most of the days after the 14th of March have many outlier values (due to low battery voltage), we decided to focus our attention on the period from the 1st to the 14th March, creating a dataset with measurements taken only within this period. We focus on temperature measurements. To avoid the fact that some sensors do not provide measurements for each time instance, we grouped measurements in a time slot of t = 15 min, and considered the average values of temperature in the considered timeslot.

The experiments of the algorithms have been performed on a dataset containing sensors 1, 2, 3, 4, 6, 7, 9, 10, 33, and 35. Their measurements are characterized by a high spatial correlation that can be exploited in the imputation process, as shown in the correlation matrix of Fig. 5.

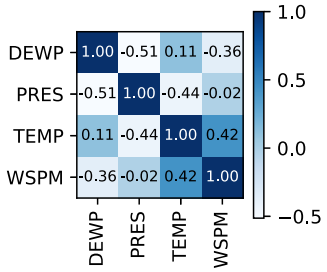


Fig. 6. Correlation matrix of the four sensors from the moderate correlation version of the AirQuality Beijing dataset.

The approaches SAITS and M-RNN are compared with GARDA using the Beijing Multisite Air-Quality dataset [58] (https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data). It includes hourly air pollutants and meteorological data from 12 monitoring sites in Beijing, between 2013/03/01 and 2017/02/28 (48 months total). Specifically, each dataset row contains a row id, date and time, concentration in μg/m^3 of PM2.5, PM10, SO2, NO2, CO, O3, and the following meteorological variables: TEMP—temperature (°C), PRES—pressure (hPa), DEWP—dew point temperature (°C), RAIN—precipitation (mm), WSPM—wind speed (m/s), WD—wind direction. Since our method focuses on continuous data, we reduced the dataset to the following variables: TEMP, PRES, DEWP, WSPM; the variable RAIN was excluded because it assumes a value of zero for a large portion of the dataset. Moreover, since our method focuses on imputing short time frames locally as they are generated by IoT devices, we limited the time period to 40 days, and we consider data from five stations (Dingling, Dongsi, Nongzhanguan, Aotizhongxin, and Guanyuan). To ensure a fair comparison, SAITS and M-RNN approaches use this time frame as the test set, while the remaining data is split into a validation set and a training set in a 1:3 ratio. The measurements are characterized by high correlation between measurements of the same sensor type across different stations. To also show a case with lower correlation, we modified the Air-Quality dataset to include data from only one station (Dingling), because most of the correlation is observed between data coming from the same sensors in different stations. The correlation matrix of this new version of the dataset is depicted in Fig. 6. To make the comparison between datasets at varying correlations as fair as possible, since the dataset with one station has a lower number of columns, we include 200 days to keep the same number of data points.

B. Evaluation Protocol

To evaluate the ability of an algorithm to properly reconstruct the data, we must have a ground truth. Therefore, starting from a complete dataset, we inject errors to remove some of the data; then, we reconstruct the data and compare the reconstructed values with the ground truth to evaluate the accuracy of the imputation method. Specifically, the experiments have been conducted using the experimental protocol depicted in Fig. 7, with an approach often used in related works [9], [59], [60]. Since real-world datasets can be affected

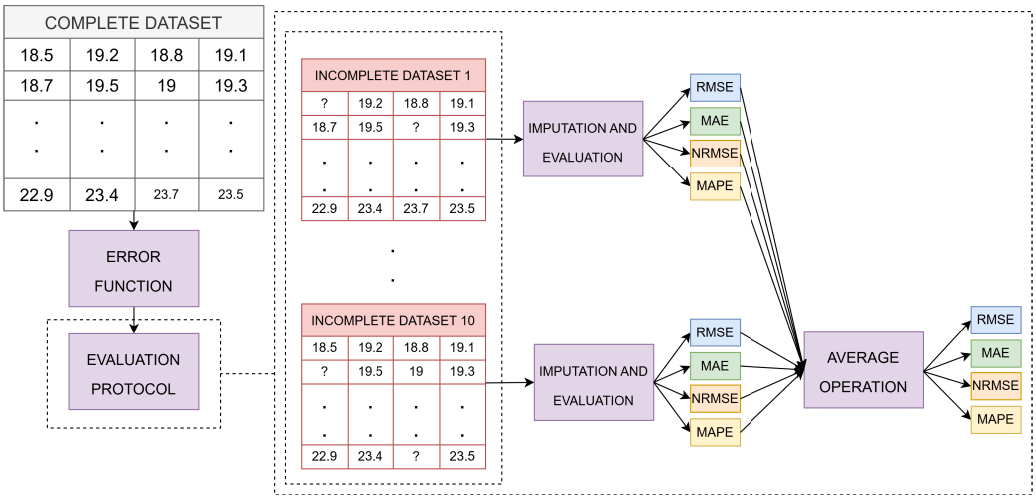


Fig. 7. Evaluation approach of GARDA on various datasets.

by different degrees of data loss, it is essential to perform the evaluation under different error severities. In particular, we produced several datasets by injecting missing data at rates of 5%, 10%, 20%, 30%, 40%, and 50% [9], [59], [60], [61]. Different error models can be employed to inject errors in a dataset. We applied the MCAR and the consecutive error model to the Intel Lab dataset. For the Beijing Multisite Air-Quality dataset, however, we focused on the MCAR model, as it is the only applied error model in the corresponding article [44], to evaluate both the proposed method and the approach [43]. For the sake of consistency with the experimental phase on the Intel Lab dataset, we decided to conduct the comparison for missingness ranging from 10% to 50%. MCAR produces completely random errors, while the consecutive error model simulates a more realistic case in which a sensor or a group of sensors does not produce any value for a certain amount of time (e.g., due to network errors, blackout, etc.): these missingness periods range from 4 to 64 h. Given that a row takes the average value of those sensors over a 15-min interval, the number of consecutive missing values of certain sensors goes from 16 to 256. To obtain reliable results and minimize the effects of random fluctuations, we generated ten different datasets using random seeds for each combination of error model and missing rate. The evaluation metrics are computed as the average on the ten different datasets for each configuration. Specifically, the following metrics are considered:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{9}$$

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{10}$$

$$NRMSE = \frac{\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}}{max(y_1, y_2, \ldots, y_N) - min(y_1, y_2, \ldots, y_N)} \tag{11}$$

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \tag{12}$$

where y_i is the ith true value and $\hat{y}_i$ the ith predicted value. Since the Air Quality dataset includes values that are close to zero, the MAPE is not a reliable metric to measure the relative error between real and imputed values. This is because the true values appear in the denominator, resulting in excessively high MAPE values when they are near zero. The MRE metric solves this problem by calculating the ratio between the sums of the errors and the true values

$$MRE = \frac{\sum_{i=1}^{N}|y_i - \hat{y}i|}{\sum_{i=1}^{N}|y_i|}. \tag{13}$$

Therefore, when comparing our method to ML approaches, we use MRE instead of MAPE.

This procedure is performed to compute the metrics for all the algorithms used in the comparison, in particular: mean, median, mode, LR, KNN, K-means, and SVD-based for the Intel Lab Dataset, while we compare against M-RNN and SAITS for the Beijing Multisite Air-Quality dataset.

C. Results and Discussion

The optimal configuration of the granular algorithm (Algorithm 1) was determined through a structured grid search. Given their significant impact on the rule-mining process, the initial minimum support ($initSup$) and initial minimum confidence ($initConf$) were defined first by testing all possible combinations within the following ranges, using a fixed step size of 0.05.

1) $initSup \in [0.35, 0.75]$.
2) $initConf \in [0.70, 0.95]$.

Following the optimization of these variables, the search was extended to the relaxation factors $rSup$ and $rConf$. The first was evaluated from 10 to 130 with a step of 30, while the latter was tested in the range $[1.0, 2.0]$ with increments of 0.25. It is important to note that $rConf$ is logically constrained to values ≥ 1, otherwise it would incorrectly increase rule confidence rather than relaxing it. The optimal configuration identified through this grid search is as follows.

1) $initSup = 0.65$.

2) $initConf = 0.9$.
3) $rSup = 100$.
4) $rConf = 1$.
5) $numMaxIter = 1000$.

Other critical aspects are the chosen granulation technique and the granule size. For our implementation of GARDA, we tested three different granulation approaches, already discussed in Section III-B, namely equal-width, K-means, and KDE.

The granule size is also critical, as it is linked both to the number of items involved in the reconstruction and to the accuracy with which the continuous value is recovered from the granule. Larger granules reduce the total number of items (leading to higher support and confidence), but they also introduce greater inaccuracy in the reconstruction of the original values because the average value of the granule is computed over a wide interval. Conversely, smaller granules increase the number of items (resulting in many items with lower support and confidence), but when the correct granule is selected, the original value can be reconstructed more accurately. For this reason, for both datasets we determined the granule size using a grid search based on the standard deviation σ computed from the dataset. In particular, we tested the following values: $\sigma, (\sigma/2), (\sigma/3), (\sigma/4), (\sigma/5), (\sigma/6), (\sigma/7)$, and$(\sigma/8)$. For the Intel Lab dataset, considering that σ calculated over the entire dataset is 2.31, the best value was obtained with K-means, with an indicative granule size of $(\sigma/6)$ (0.385), corresponding to 41 clusters. Obviously, in the case of K-means, this is only an indicative value used in the grid search, as the algorithm adjusts the size of the various clusters based on the data. Similarly, K-means provided the best performance also for the Beijing dataset; the grid search was performed by starting from a fraction of σ considered reasonable for the scale of the values in the column, then progressively reducing it until reaching the first value at which performance deteriorates. The fraction of σ immediately preceding this value is then taken as the optimal choice. Following such procedure, for TEMP ($\sigma = 4.75$), we tested $(\sigma/4), (\sigma/5), (\sigma/6), (\sigma/7), (\sigma/8)$, and the best value was $(\sigma/7)$ (0.679), corresponding to 42 clusters. For PRES ($\sigma = 7.52$), we tested $(\sigma/12), (\sigma/13), (\sigma/14), (\sigma/15), (\sigma/16)$, and the best value was $(\sigma/15)$ (0.501), corresponding to 88 clusters. For DEWP ($\sigma = 6.35$), we tested $(\sigma/6), (\sigma/7), (\sigma/8), (\sigma/9), (\sigma/10)$, and the best value was $(\sigma/9)$ (0.706), corresponding to 46 clusters. For WSPM ($\sigma = 1.55$), we tested $\sigma, (\sigma/2), (\sigma/3), (\sigma/4)$, and the best value was $(\sigma/3)$ (0.517), corresponding to 18 clusters.

For the traditional data imputation algorithm (second phase of the GARDA architecture in Fig. 1), a Bayesian ridge regression model with the following parameters is used.

1) $\alpha_1 = 1e^{-6}$*:* Shape parameter for the gamma distribution prior over the alpha parameter.
2) $\alpha_2 = 1e^{-6}$*:* Inverse scale parameter (rate parameter) for the gamma distribution prior over the alpha parameter.
3) $\lambda_1 = 1e^{-6}$*:* Shape parameter for the gamma distribution prior over the lambda parameter.
4) $\lambda_2 = 1e^{-6}$*:* Inverse scale parameter (rate parameter) for the gamma distribution prior over the lambda parameter.
5) $\alpha_{init} = (1/Var(Y))$*:* Initial value for alpha.
6) $\lambda_{init} = 1$*:* Initial value for lambda.

These are the parameters of the other imputation algorithms compared with GARDA: mean, median, and mode approaches have no parameters to tune; the LR has the same parameters used in our method; KNN has a number (K) of neighbors of 10; K-means has a K value of 20; the SVD-based approach has a shrinkage value equal to max(singular value init matrix)/50. These values have been tuned with a grid-search approach to find the best parameters for the considered datasets. The hyperparameters for the ML methods are the same used by the SAITS article [44], as they have already been optimized for the Air Quality dataset. They are set to the following.

1) *SAITS and M-RNN:* Sequence length 24, batch size 128, ADAM optimizer with 0 weight decay, early stop patience 30.
2) *SAITS Only:* Learning rate 0.000882, hidden dim 512, hidden size of feed forward layer 512, self-attention number of heads 4, key dim 128, value dim 64, dropout rate 0, inner group parameter sharing with 1 group.
3) *M-RNN Only:* Learning rate 0.000924, hidden size 256.

Tables I and II report the means and standard deviations obtained on the ten datasets, for all the metrics. These tables do not contain mean, median, and mode imputation results because their performance is far worse than the others (at least by one order of magnitude). The KNN is the algorithm with the lowest RMSE for missing rates below 40%, while its performance strongly decays for missing rates above 30%. On the other hand, GARDA has the lowest RMSE for high missing rates, especially for the 40% rate. Indeed, GARDA improves upon LR for high missing rates, with an improvement of about 20% for both a 40% and a 50% missing rate. Looking at the MAE, the KNN is the best for low missing rates, while our approach has the lowest MAE for 40% and 50% missing rates. In terms of MAE, GARDA improves upon LR by 15% for an error rate of 50%. To be more precise, our approach achieves an average absolute error of approximately 0.29 C°–0.33 C° on 40% and 50% datasets against 0.33 C°–0.39 C° of the best algorithm (LR-only). The performance of GARDA has also been evaluated under a consecutive error model using the Intel Lab Dataset (Tables I and II). Under this model, KNN performs worse than in the case of random missingness.

The best-performing model is GARDA, as it matches LR, which is the traditional model with the best performance, at a 5% missing rate and outperforms all the other methods in imputing the datasets from 10% upward. This advantage becomes much more evident at higher missing rates. Specifically, GARDA outperforms LR with RMSE reductions of 46% at 10% missingness (0.378 versus 0.700), 52% at 20% (0.529 versus 1.109), 62% at 30% (0.642 versus 1.715), 70% at 40% (0.752 versus 2.529), and 67% at 50% (1.206 versus 3.761). When considering MAE, GARDA matches LR at 5% and outperforms all other methods at 10%, 20%, 30%, 40%, and 50% missing rates, leading, for example, to a 71% MAE reduction at a 50% missing rate with respect to LR.

Tables I and II show the performance of the Hong and Wu [42] approach, in which all missing values are imputed using granular association rules instead of leaving the remaining

TABLE I

TEST RESULTS FROM THE MISSING COMPLETELY RANDOM ERROR MODEL

Rate	Measure	KNN				LR				SVD				KMEANS				HONG et al. [42]				GARDA			
		rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape
5%	μ	**0.199**	**0.101**	**0.013**	**0.004**	0.301	0.189	0.019	0.008	0.456	0.282	0.029	0.012	0.516	0.367	0.033	0.016	2.274	1.608	0.145	0.071	0.299	0.190	0.019	0.008
	σ	**0.0201**	**0.0073**	**0.0013**	**0.0003**	0.0205	0.0096	0.0013	0.0004	0.0271	0.0010	0.0017	0.0004	0.0270	0.0140	0.0017	0.0006	0.1958	0.1026	0.0125	0.0037	0.017	0.0079	0.0011	0.0004
10%	μ	**0.220**	**0.111**	**0.014**	**0.005**	0.326	0.202	0.021	0.009	0.488	0.297	0.031	0.013	0.528	0.372	0.034	0.017	2.348	1.660	0.150	0.073	0.320	0.200	0.020	0.009
	σ	**0.0021**	**0.0070**	**0.0014**	**0.0003**	0.0014	0.0083	0.0009	0.0004	0.0151	0.0006	0.0010	0.0003	0.0252	0.0099	0.0016	0.0004	0.2062	0.1260	0.0132	0.0055	0.0148	0.0074	0.0009	0.0003
20%	μ	**0.273**	**0.137**	**0.017**	**0.006**	0.378	0.227	0.024	0.010	0.523	0.317	0.033	0.014	0.534	0.380	0.034	0.017	2.153	1.535	0.137	0.068	0.353	0.218	0.023	0.010
	σ	**0.0019**	**0.0054**	**0.0012**	**0.0002**	0.0014	0.0007	0.0009	0.0003	0.0172	0.0063	0.0011	0.0003	0.0155	0.0010	0.0100	0.0004	0.1329	0.0869	0.0085	0.0032	0.0108	0.0059	0.0007	0.0003
30%	μ	**0.364**	**0.188**	**0.023**	**0.008**	0.470	0.272	0.030	0.012	0.580	0.350	0.037	0.015	0.551	0.385	0.035	0.017	2.220	1.579	0.142	0.069	0.422	0.258	0.027	0.011
	σ	**0.0135**	**0.0051**	**0.0009**	**0.0002**	0.0030	0.0073	0.0019	0.0003	0.0241	0.0075	0.0015	0.0003	0.0153	0.0057	0.0010	0.0003	0.1783	0.1097	0.0144	0.0046	0.0195	0.0085	0.0012	0.0004
40%	μ	0.575	0.306	0.037	0.013	0.583	0.330	0.037	0.015	0.642	0.384	0.041	0.017	0.576	0.401	0.037	0.018	2.122	1.494	0.135	0.066	**0.478**	**0.289**	**0.031**	**0.013**
	σ	0.0331	0.0146	0.0021	0.0006	0.0027	0.0086	0.0017	0.0004	0.0169	0.0039	0.0011	0.0002	0.0150	0.0086	0.0010	0.0004	0.1799	0.1438	0.0115	0.0060	**0.0167**	**0.0091**	**0.0011**	**0.0004**
50%	μ	0.841	0.484	0.054	0.021	0.698	0.398	0.044	0.018	0.718	0.433	0.046	0.019	0.610	0.420	0.039	0.019	1.956	1.333	0.125	0.066	**0.568**	**0.333**	**0.036**	**0.015**
	σ	0.0222	0.0112	0.0014	0.0005	0.0042	0.0017	0.0027	0.0007	0.0105	0.0041	0.0007	0.0002	0.0263	0.0098	0.0017	0.0004	0.0688	0.0469	0.0044	0.0021	**0.0376**	**0.0134**	**0.0024**	**0.0006**

TABLE II

TEST RESULTS FROM THE CONSECUTIVE ERROR MODEL

Rate	Measure	KNN				LR				SVD				KMEANS				HONG et al. [42]				GARDA			
		rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape
5%	μ	0.404	0.266	0.026	0.012	**0.339**	**0.218**	**0.022**	**0.010**	0.423	0.298	0.027	0.013	1.086	0.664	0.069	0.028	2.178	1.703	0.139	0.076	0.345	0.218	0.022	0.010
	σ	0.0853	0.0561	0.0054	0.0025	**0.0880**	**0.0500**	**0.0056**	**0.0022**	0.1101	0.0686	0.0070	0.0029	0.4299	0.2142	0.0274	0.0086	0.4670	0.3733	0.0298	0.0159	0.0934	0.0564	0.0060	0.0024
10%	μ	0.947	0.595	0.060	0.027	0.700	0.453	0.045	0.020	0.896	0.622	0.057	0.027	2.359	1.403	0.151	0.060	2.381	1.817	0.152	0.081	**0.378**	**0.240**	**0.024**	**0.011**
	σ	0.2283	0.0976	0.0146	0.0041	0.0682	0.0495	0.0044	0.0023	0.1249	0.0679	0.0080	0.0027	0.5328	0.2316	0.0340	0.0087	0.6254	0.4347	0.0399	0.0182	**0.0707**	**0.0522**	**0.0045**	**0.0024**
20%	μ	1.678	1.002	0.107	0.044	1.109	0.716	0.071	0.032	1.529	1.014	0.098	0.044	3.805	2.217	0.243	0.095	2.511	1.833	0.160	0.079	**0.529**	**0.300**	**0.034**	**0.013**
	σ	0.2839	0.1061	0.0181	0.0040	0.0560	0.0392	0.0036	0.0017	0.2119	0.1016	0.0135	0.0039	0.3889	0.1846	0.0248	0.0071	0.4275	0.3320	0.0273	0.0145	**0.2166**	**0.0720**	**0.0138**	**0.0031**
30%	μ	2.576	1.474	0.164	0.065	1.715	1.068	0.109	0.047	2.277	1.458	0.145	0.063	5.326	3.070	0.340	0.131	2.519	1.784	0.161	0.078	**0.642**	**0.362**	**0.041**	**0.016**
	σ	0.2539	0.0878	0.0162	0.0033	0.1573	0.0682	0.0100	0.0030	0.2031	0.0889	0.0130	0.0033	0.3317	0.1525	0.0212	0.0057	0.3971	0.2427	0.0253	0.0106	**0.1762**	**0.0672**	**0.0112**	**0.0026**
40%	μ	3.575	2.019	0.228	0.088	2.529	1.555	0.161	0.068	3.129	1.975	0.200	0.085	6.928	4.007	0.442	0.171	2.330	1.595	0.149	0.069	**0.752**	**0.433**	**0.048**	**0.019**
	σ	0.1746	0.0564	0.0111	0.0019	0.3110	0.1560	0.0198	0.0068	0.1659	0.0811	0.0106	0.0033	0.2557	0.0966	0.0163	0.0036	0.3466	0.2278	0.0221	0.0098	**0.1356**	**0.0620**	**0.0087**	**0.0025**
50%	μ	5.054	2.791	0.323	0.120	3.761	2.293	0.240	0.100	4.190	2.617	0.267	0.112	8.826	5.139	0.563	0.220	2.381	1.542	0.152	0.066	**1.206**	**0.649**	**0.077**	**0.028**
	σ	0.2259	0.1137	0.0144	0.0050	0.3318	0.1721	0.0212	0.0075	0.1857	0.0929	0.0119	0.0039	0.2052	0.1063	0.0131	0.0047	0.3297	0.2473	0.0210	0.0104	**0.2754**	**0.0953**	**0.0176**	**0.0035**

TABLE III

GRANULATION METHOD TEST RESULTS FROM THE CONSECUTIVE ERROR MODEL

Rate	Measure	EQUAL-WIDTH (41 intervals)				KMEANS (41 clusters) [55]				KERNEL DENSITY [56]			
		rmse	mae	nrmse	mape	rmse	mae	nrmse	mape	rmse	mae	nrmse	mape
5%	μ	0.348	0.222	0.022	0.0010	**0.345**	**0.218**	**0.022**	**0.010**	0.915	0.571	0.058	0.026
	σ	0.0895	0.0529	0.0057	0.0023	**0.0934**	**0.0564**	**0.0060**	**0.0024**	0.2230	0.1599	0.0142	0.0072
10%	μ	0.379	0.242	0.024	0.011	**0.378**	**0.240**	**0.024**	**0.011**	0.947	0.586	0.060	0.026
	σ	0.0692	0.0515	0.0044	0.0023	**0.0707**	**0.0522**	**0.0045**	**0.0024**	0.1457	0.1127	0.0093	0.0047
20%	μ	**0.522**	0.300	**0.033**	0.013	0.529	**0.300**	0.034	**0.013**	1.125	0.720	0.072	0.032
	σ	**0.3052**	0.1149	**0.0195**	0.0045	0.2166	**0.0720**	0.0138	**0.0031**	0.1248	0.0852	0.0080	0.0036
30%	μ	0.681	0.374	0.043	0.016	**0.642**	**0.362**	**0.041**	**0.016**	1.210	0.786	0.077	0.035
	σ	0.2523	0.0873	0.0161	0.0034	**0.1762**	**0.0672**	**0.0112**	**0.0026**	0.1427	0.0884	0.0091	0.0036
40%	μ	0.856	0.464	0.055	0.020	**0.752**	**0.433**	**0.048**	**0.019**	1.440	0.948	0.092	0.041
	σ	0.1923	0.0756	0.0123	0.0029	**0.1356**	**0.0620**	**0.0087**	**0.0025**	0.2339	0.1263	0.0149	0.0051
50%	μ	1.284	0.676	0.082	0.028	**1.206**	**0.649**	**0.077**	**0.028**	1.784	1.139	0.114	0.049
	σ	0.3221	0.1147	0.0206	0.0044	**0.2754**	**0.0953**	**0.0176**	**0.0035**	0.2448	0.1265	0.0156	0.0053

missing values to the traditional method. The poor results are due to the fact that missing values imputed using rules with low confidence and support lead to very large errors, highlighting the importance of a hybrid approach.

A summary of the granulation method grid search conducted on the Intel Lab dataset, under the consecutive error model, is shown in Table III, which compares the best configurations identified for each granulation approach. K-means slightly outperforms equal-width at low missing rates (5% to 20%), with an average reduction in RMSE and MAE of 2%, except for RMSE and NRMSE at 20% missing rate. Greater improvements can be observed at higher missing rates: at 30% missingness, RMSE and MAE decrease by 5% and 3%, respectively; at 40%, by 12% and 7%; and at 50%, by 6% and 4%.

Table IV reports the metrics calculated on the Beijing Air Quality dataset, specifically the version with high correlation (five stations). SAITS shows better performances across all metrics at 5% and 10% missing rate. At 20% missing rate and above, GARDA outperforms SAITS, reaching a 29% reduction in RMSE (0.215 versus 0.301) and 34% in MAE (0.083 versus 0.125) at 50% missing rate.

Finally, Table V includes the metrics calculated on the moderate correlation version of the dataset (one station). SAITS demonstrates better results across all metrics and all missing rates, showing the capability of extracting complex patterns

TABLE IV
TEST RESULTS AGAINST NN-BASED METHODS—HIGH CORRELATION VERSION

Rate	GARDA			M-RNN			SAITS		
	RMSE	MAE	MRE	RMSE	MAE	MRE	RMSE	MAE	MRE
5	0.191	0.076	0.094	0.404	0.259	0.374	**0.167**	**0.055**	**0.080**
10	**0.185**	0.074	**0.091**	0.450	0.285	0.404	0.185	**0.065**	0.093
20	**0.183**	**0.074**	**0.091**	0.492	0.330	0.473	0.229	0.083	0.119
30	**0.190**	**0.076**	**0.093**	0.514	0.350	0.502	0.258	0.100	0.143
40	**0.195**	**0.078**	**0.095**	0.541	0.375	0.537	0.278	0.110	0.158
50	**0.215**	**0.083**	**0.101**	0.574	0.400	0.573	0.301	0.125	0.179

TABLE V
TEST RESULTS AGAINST NN-BASED METHODS—MODERATE CORRELATION VERSION

Rate	GARDA			M-RNN			SAITS		
	RMSE	MAE	MRE	RMSE	MAE	MRE	RMSE	MAE	MRE
5	0.685	0.521	0.641	0.774	0.626	0.689	**0.333**	**0.167**	**0.188**
10	0.703	0.534	0.659	0.810	0.654	0.731	**0.323**	**0.172**	**0.193**
20	0.726	0.554	0.686	0.840	0.680	0.766	**0.333**	**0.172**	**0.194**
30	0.756	0.579	0.716	0.844	0.715	0.805	**0.345**	**0.181**	**0.203**
40	0.784	0.602	0.745	0.905	0.737	0.833	**0.350**	**0.182**	**0.205**
50	0.812	0.626	0.776	0.933	0.763	0.86	**0.368**	**0.199**	**0.224**

when the correlation is lower. However, GARDA performs better in all metrics and missing rates against M-RNN, probably because the latter relies even more on high correlation. In addition, it should be considered that GARDA maintains good performance and can be run even on devices with low computational power and battery, unlike NN approaches that need GPUs and higher energy loads.

D. Discussion

The experimental phase shows promising results in terms of imputation, being capable of outperforming both traditional and machine learning approaches. The experiments demonstrate distinct advantages and disadvantages, highlighting the situations in which the method performs best. Specifically, GARDA shows significant improvements at missing rates higher than 10% when the dataset exhibits high correlation. In fact, the GARDA approach is designed to be integrated into IoT-based systems strongly focused on edge computing, where small devices with low power, battery, and computational and storage capabilities work on small IoT networks, with highly correlated sensors.

The GARDA performances reported in Tables I, II, IV, and V have been obtained by integrating K-means as the granulation method, as a grid search identified it as the top-performing approach compared to equal-width and KDE. The grid search on the Intel Lab dataset shows that K-means consistently performs better across all metrics and missing rates, with the exception of a slight difference in RMSE and NRMSE compared to equal-width at the 20% missing rate. This behavior can be explained by the greater flexibility of K-means in determining the size of the granules, unlike equal-width, where the size is fixed. This flexibility allows K-means to produce narrower clusters in denser areas and wider clusters in sparser areas. In this way, the rules with the highest confidence correspond to more frequent values and narrower intervals, which are reconstructed more accurately.

Furthermore, GARDA's improvements for MCAR models over traditional methods at high missing rates effectively reflect what we expected. The idea behind GARDA originated from the observation that traditional methods are computationally lightweight but tend to perform poorly when the missing rate is high. Our aim was therefore to develop a hybrid approach that combines rule-based and traditional techniques to enhance performance. In small, highly correlated sensor networks, rule-mining approaches are able to extract meaningful rules that can be effectively used for imputation. From our results, we observed that the method improves upon traditional approaches because it partially imputes the dataset using the best extracted rules, producing values that outperform those obtained by traditional methods alone. Once the dataset has been partially imputed with high-quality values, traditional methods can operate under more favorable conditions and achieve better imputations when combined with GARDA. This aspect is fundamental in explaining the performance improvements obtained over the other approaches. In these cases, it is even possible to achieve results that surpass those of NN-based methods. However, this advantage diminishes at low missing rates, because traditional methods already perform well under such conditions and often outperform rule-mining approaches in filling most of the gaps.

However, our method has shown that, in the context of the consecutive error model with highly correlated sensors, when there are many consecutive gaps (for instance, when sensors malfunction and do not produce values for long periods of time), it performs better than other traditional methods, and even significantly better when the missing rate is high. This is also due to the difficulty traditional methods have in performing well under such conditions. As shown in Fig. 8, which compares the true values with the ones reconstructed by GARDA for four example sensors at a 50% missing rate, the trend of the original values is accurately captured, with only small fluctuations around the true values. A noticeable deviation between the curves can be observed only in the presence of atypical spikes in the data, as in the case of temperature sensor 35. Therefore, the usefulness of GARDA in this context is even greater, as it is able to support traditional methods more effectively.

The effectiveness of the proposed hybrid approach is inherently tied to the presence of correlations within the dataset. In scenarios characterized by high intersensor correlation, the GrC Association-rule-based imputation builds a dense set of high-confidence association rules. However, as the degree of correlation decreases, the number of effective rules that can be mined diminishes. In these instances, the hybrid algorithm is more reliant on the traditional method (in our case, LR). Another critical factor in the resulting performance degradation is the sequential nature of the hybrid process. The traditional method is applied to a dataset that has already been partially imputed by the granular association rules. When the available rules are weak, the initial imputation phase provides a lower-quality data foundation, which can lead to a propagation of errors and a further decline in overall accuracy after the traditional method is applied. Despite these limitations, the simplicity of the model and its effectiveness on small, highly correlated datasets are very valuable in the context of IoT systems and edge computing. Given that sensor networks

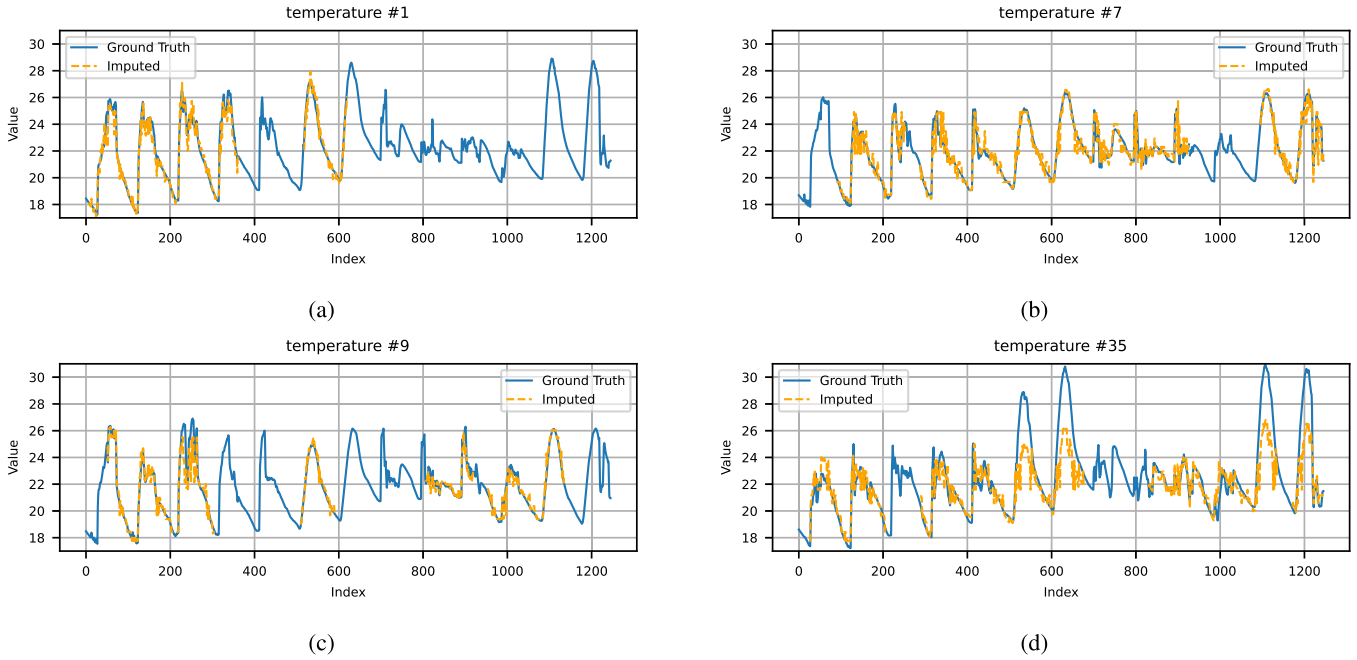


Fig. 8. Comparison between the real and imputed values for four variables of the Intel Lab dataset at 50% missing rate under the consecutive error model. (a) Sensor 1 of Intel Lab dataset, (b) sensor 7, (c) sensor 9, and (d) sensor 35.

in industrial or environmental monitoring typically exhibit strong spatial and temporal dependencies, the assumption of correlation is physically grounded. This makes our hybrid approach a suitable candidate for resource-constrained devices, where more resource-intensive techniques are unfeasible.

V. Conclusion

Data missingness is a major challenge in IoT systems. In this article, we proposed GARDA, a hybrid imputation approach based on granular association rules and traditional imputation methods, for small and highly correlated datasets. The method granulates sensor data into intervals, extracts association rules among granules, and uses them to recover missing values, while any remaining missing data are handled by a traditional imputation technique.

GARDA was compared with several imputation algorithms on the Intel Lab Dataset and with two state-of-the-art machine learning methods on two versions of the Air Quality Beijing dataset. Results show that GARDA performs well under different error models and missing rates, outperforming traditional methods at high missing rates and both machine learning methods at moderate and high missing rates in highly correlated settings.

As future directions, we will explore other granulation approaches that could take into account uncertainty and vagueness in data (e.g., fuzzy sets and rough sets), as well as different reconstruction schemes beyond the midpoint operator or the cluster centroid, to further improve performance. Another useful extension is the explicit adaptation of GARDA to streaming-data scenarios. Currently, the method is designed for settings in which the available data are first collected and then processed as a whole to infer granules and association rules. However, the same approach can be adapted to streaming data by progressively updating both components as new observations become available. This evolution would allow the model to adapt to environmental variations, seasonality, sensor aging, or evolving operating conditions. In particular, a streaming-oriented version of GARDA could rely on sliding or adaptive windows to progressively refresh granules and association rules based on recent data, thereby updating the imputation model without fully recomputing it. Finally, the proposed approach will also be evaluated in other application domains.

Credit Authorship Contribution Statement

Giuseppe D'Aniello—Conceptualization, methodology, validation, formal analysis, investigation, data curation, visualization, supervision, writing—original draft, and writing—review, and editing.

Mario Della Corte—Software, validation, formal analysis, investigation, data curation, visualization, and writing—review and editing.

Rosario Gaeta—Conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, supervision, writing—original draft, and writing—review and editing.

Tzung-Pei Hong—Methodology, validation, formal analysis, investigation, supervision, writing—original draft, and writing—review and editing.

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Open Access funding provided by 'Universita degli Studi di Salerno' within the CRUI CARE Agreement